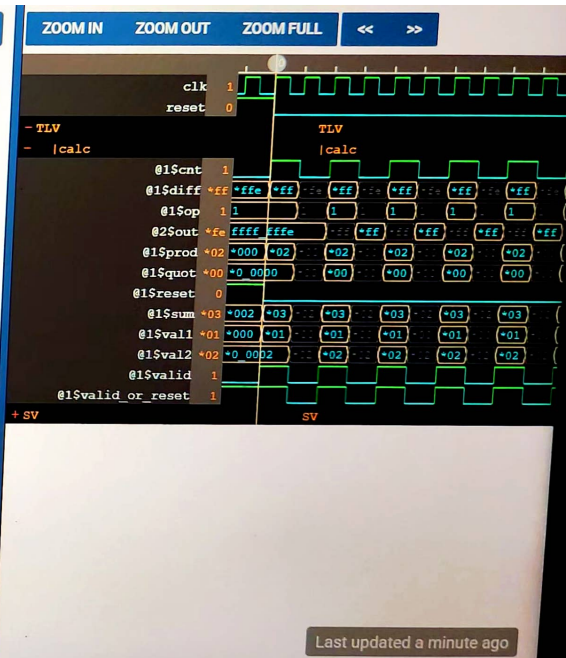
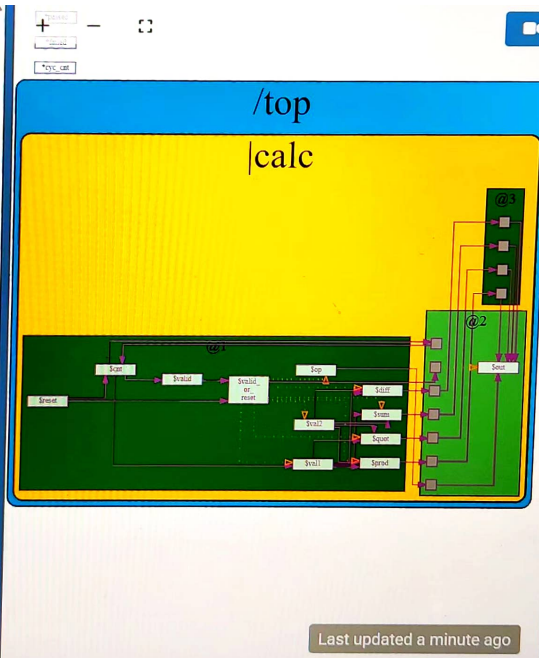


```

14 >calc
15 @1
16 $reset = *reset;
17 $cnt = $reset ? 1'b0 : ~>>1$cnt;
18
19 $valid = $cnt;
20
21 $valid_or_reset = $valid || $reset;
22
23 ?$valid_or_reset I
24   $val1[31:0] = $cnt;
25   $val2[31:0] = 32'd2;
26   $op[1:0]    = 2'b01;
27
28   $sum[31:0] = $val1 + $val2;
29   $diff[31:0] = $val1 - $val2;
30   $prod[31:0] = $val1 * $val2;
31   $quot[31:0] = $val2 != 0 ? $val1 /
32
33 @2
34
35 ?$valid_or_reset
36   $out[31:0] = $op == 2'b00 ? >>1$sum
37   $op == 2'b01 ? >>1$di
38   $op == 2'b10 ? >>1$pr
39   >>1$qu
40

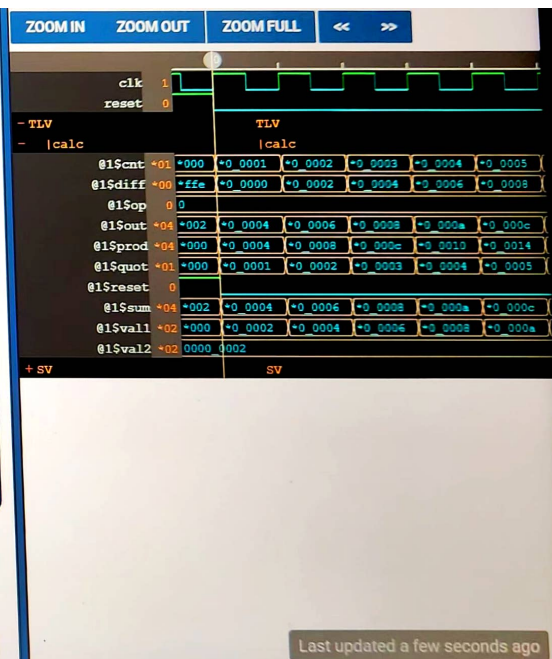
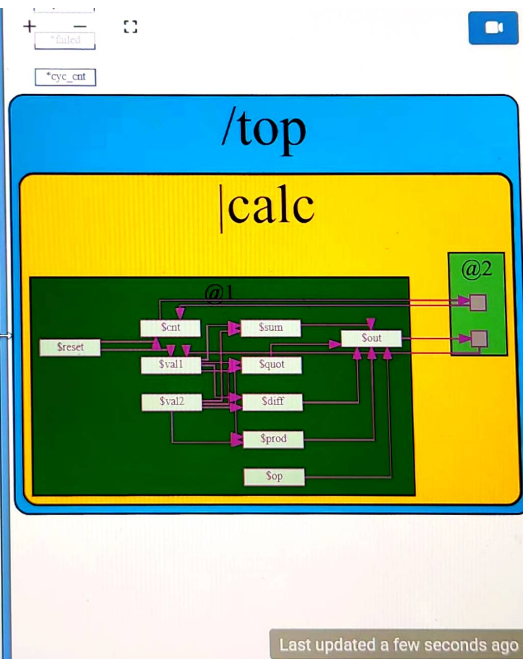
```



```

12 _makerchip_module // (Expanded in Nav E
13
14 *alc
15 @1
16 $reset = *reset;
17 Scnt[31:0] = $reset ? 32'd0 : >>1$scnt + 1
18
19 $val1[31:0] = $reset ? 0 : >>1$scnt;
20 $val2[31:0] = 32'd2;
21 $op[1:0] = 2'b00;
22
23 $sum[31:0] = $val1 + $val2;
24 $diff[31:0] = $val1 - $val2;
25 $prod[31:0] = $val1 * $val2;
26 $quot[31:0] = $val2 != 0 ? $val1 / $val2
27
28 $out[31:0] = $op == 2'b00 ? $sum :
29 $op == 2'b01 ? $diff :
30 $op == 2'b10 ? $prod :
31 $quot;
32
33 ased = *cyc_cnt > 40;
34 ailed = 1'b0;
35
36 dmodule
37

```



makerchip.com/ide/

PROJECT LEARN IDE

EDITOR NAV-TLV LOG

```
10 // Macro providing required top-level
11 // stimulus support, and Verilator config.
12 m5_makerchip_module // (Expanded in Nav-TL
13
14 TLV
15 // Connect SV inputs to TLV pipesignals.
16 $reset = *reset;
17
18 $val1[3:0] = 4'd8;
19 $val2[3:0] = 4'd2;
20 $op[1:0] = 2'b00;
21
22 $sum[31:0] = $val1 + $val2;
23 $diff[31:0] = $val1 - $val2;
24 $prod[31:0] = $val1 * $val2;
25 $quot[31:0] = $val1 / $val2;
26
27 Sout[31:0] =
28   $op == 2'b00 ? $sum :
29   $op == 2'b01 ? $diff :
30   $op == 2'b10 ? $prod :
31   $op == 2'b11 ? $quot :
32   0;
33
34 *passed = *cyc_cnt > 40;
35 *failed = 1'b0;
36
37 // Assert these to end simulation (before tl
```

DIAGRAM VIZ

reset

*passed

*failed

*cyc_cnt

/top

@0

Sreset

Sop

Sval1

Sval2

Ssum

Sdiff

Squot

Sprod

Sout

WAVEFORM

ZOOM IN ZOOM OUT ZOOM FULL << >>

clk

reset

TLV

00\$diff 0000 0000

00\$op 0 0

00\$out 0000 0000

00\$prod 0000 0100

00\$quot 0000 0000

00\$reset 0

00\$sum 0000 0000

00\$val1 8 8

00\$val2 2 2

SV

SV

Last updated a few seconds ago

Last updated a few seconds ago

EDITOR

NAV-TLV

LOG

14 **alc**

15 **@1**

16 \$reset = *reset;

17 \$cnt = \$reset ? 1'b0 : ~>1\$cnt;

18

19 \$valid = \$cnt;

20

21 \$valid_or_reset = \$valid || \$reset;

22

23 **?\$valid_or_reset**

24 \$val1[31:0] = \$cnt;

25 \$val2[31:0] = 32'd2;

26 \$op[1:0] = 2'b01;

27

28 \$sum[31:0] = \$val1 + \$val2;

29 \$diff[31:0] = \$val1 - \$val2;

30 \$prod[31:0] = \$val1 * \$val2;

31 \$quot[31:0] = \$val2 != 0 ? \$val1 / \$val2

32

33 **@2**

34

35 **?\$valid_or_reset**

36 \$out[31:0] = \$op == 2'b00 ? >>1\$sum :

37 \$op == 2'b01 ? >>1\$diff :

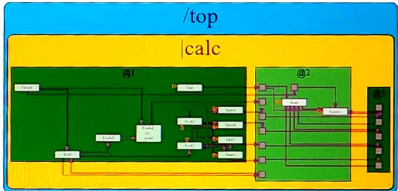
38 \$op == 2'b10 ? >>1\$prod :

39 >>1\$quot ;

40

DIAGRAM

VIZ



Last updated a few seconds ago

WAVEFORM

ZOOM IN

ZOOM OUT

ZOOM FULL

<<

>>

clk

reset

TLV

calc

01\$cnt

01\$diff

02\$mem

01\$op

02\$out

01\$prod

01\$quot

01\$reset

01\$sum

01\$val1

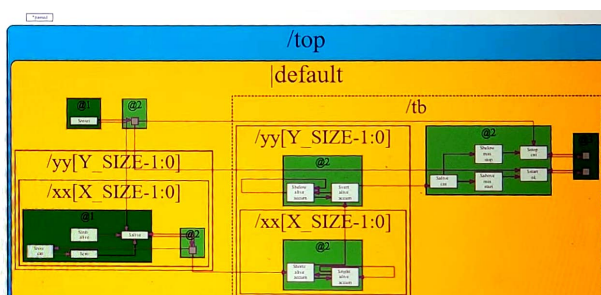


Last updated a few seconds ago

```

42
43 default
44   @1
45     $reset = *reset;
46     /yy[Y_SIZE-1:0]
47     /xx[X_SIZE-1:0]
48     @1
49     // Cell logic
50
51     // =====
52     // Population count ($cnt)
53

```



Last updated a minute ago

